SC7 Bayes Methods MT26

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Lecture 3: (a) Marginal consistency
(b) Exchangeability and its implications

Notation

If $X_1, ..., X_n$ are discrete with PMF $p_{1:n}(x_1, ..., x_n)$ on $\mathcal{X}^n$ then

$$P(X_1 = x_1, ..., X_n = x_n) = p_{1:n}(x_1, ..., x_n).$$

Subscript $1:n$ on $p_{1:n}$ gives variables and order so for $\sigma \in \mathcal{P}_{[n]}$,

$$P(X_{\sigma_1} = x_1, ..., X_{\sigma_n} = x_n) = p_\sigma(x_1, ..., x_n)$$

and

$$P(X_1 = x_{\sigma_1}, ..., X_n = x_{\sigma_n}) = p_{1:n}(x_{\sigma_1}, ..., x_{\sigma_n}).$$

If $x = (x_1, \ldots, x_n)$ and

$$P(X_1 = x_1, ..., X_n = x_n) = P(Y_1 = x_1, ..., Y_n = x_n)$$

for all $x \in \mathcal{X}^n$ then we write

$$(X_1, ..., X_n) \sim (Y_1, ..., Y_n)$$

meaning $(X_1, ..., X_n)$ and $(Y_1, ..., Y_n)$ have the same distribution.

Marginal Consistency (preamble for exchangeability)

Let $[n] = \{1, \ldots, n\}$ and $\mathcal{S}_{[n]}$ be the set of all subsets of $[n]$ excluding $\emptyset$.

For each $s \in \mathcal{S}_{[n]}$ let $r = [n] \setminus s$ with $r = (r_1, \ldots, r_m)$ say and let $p_s : \mathcal{X}^{n-m} \to [0, 1]$ be a PMF.

The family $\mathcal{F} = \{p_s,\ s \in S_{[n]}\}$ of PMFs is marginally consistent if the marginals

$$q_s(x_s) = \sum_{x_{r_1} \in \mathcal{X}} \cdots \sum_{x_{r_m} \in \mathcal{X}} p_{1:n}(x_1, \ldots, x_n)$$

satisfy $q_s(x_s) = p_s(x_s)$ for all $s \in \mathcal{S}_{[n]}$ and $x_s \in \mathcal{X}^{n-m}$.

The family is closed under taking marginals (AKA projective).

When do we *not want* marginal consistency in our prior models?

Example: students with the same birthday as Bruno de Finetti and students asking a question in a lecture.

Class with students $\{1, \ldots, n\}$ and class with just student $\{1\}$.

Let $X = (X_1, \ldots, X_n)$ with $X_i \in \{0, 1\}$ and $X \sim p_{1:n}$.

Let $Y \in \{0, 1\}$ with $Y \sim p_1$.

If $X_i = 1/0$ if student $i \in [n]$ has/not birthday on 13th June then we want $Y \sim X_1$ in our priors for the two classes*.

If $X_i = 1/0$ if student $i \in [n]$ asks/doesn't a question in class then requiring $Y \sim X_1$ in our priors for the two classes doesn't match my experience with students!

The point is that if we model parameters with a prior, and we write down a family of priors $p_s,\ s \in \mathcal{S}_{[n]}$, we should check MC.

*(ie MC with $r = 2 : n$ and $s = 1$)

Marginal Consistency (examples)

Let $x \in \{0,1\}^n$ and $k(x) = \sum_{i=1}^n x_i$. The distributions

$$p_{1:n}(x) = c_n 2^{-k(x)(n-k(x))}, \tag{1}$$

(with $c_3 = 2/7$ and $c_4 = 8/27$) are not marginally consistent as

$$p_{1:3}(0,0,0) \neq p_{1:4}(0,0,0,0) + p_{1:4}(0,0,0,1)$$

as $p_{1:3}(0,0,0) = 2/7$ and $p_{1:4}(0,0,0,0) + p_{1:4}(0,0,0,1) = 1/3$.

By contrast

$$p_{1:n}(x) = \frac{k(x)!(n-k(x))!}{(n+1)!}, \quad n \geq 1 \tag{2}$$

is marginally consistent. If $\theta \sim U(0,1)$ and $X_i \sim \text{Bern}(\theta)$ iid for $i = 1, \ldots, n$ then

$$p_{1:n}(x) = \int_0^1 \prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i} d\theta. \tag{3}$$

and we can knock out any element: $\sum_{x_1} p_{1:n}(x) = p_{2:n}(x_2, \ldots, x_n)$.

Infinite Exchangeable Sequence (IES)

Random variables $\{X_i\}_{i=1}^{\infty}$ are an IES if

$$(X_1, X_2, ..., X_n) \sim (X_{\sigma_1}, X_{\sigma_2}, ..., X_{\sigma_n})$$

for each $n \geq 1$ and all $\sigma \in \mathcal{P}_{[n]}$. Equivalently,

$$p_{1:n}(x_1, \ldots, x_n) = p_{1:n}(x_{\sigma_1}, \ldots, x_{\sigma_n})$$

for all $x \in \mathcal{X}^n$ and $\sigma \in \mathcal{P}_{[n]}$.

Example: a conditionally independent sequence of binary random variables $\theta \sim \pi(\cdot)$, $X_i \sim \text{Bern}(\theta)$, $i = 1, ..., n$ iid given θ is exchangeable as

$$p(x_1, ..., x_n) = \int_0^1 \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i}\pi(\theta)d\theta.$$

Permuting indices just shuffles the product.

Question: Taking $\theta \sim U(0,1)$ we get (3) so there is an IES with marginals in (2). Does an IES with marginals given by (1) exist?

Theorem (de Finetti): Let $X_1, X_2, ...X_n, ...$ be an infinite sequence of binary random variables. The sequence is exchangeable if and only if there exists a distribution function $F(\theta)$ on $[0, 1]$ such that

$$p(x_1, ..., x_n) = \int_0^1 \left[\prod_{i=1}^{n} p(x_i|\theta) \right] dF(\theta) \qquad (*)$$

with $F(\theta) = \Pr(\Theta \leq \theta)$ where

$$\Theta = \lim_{N \to \infty} N^{-1} \sum_{i=1}^{N} X_i \qquad \text{almost surely.}$$

It further holds that the conditional distribution is Bernoulli,

$$p(x_1, ..., x_n|\Theta = \theta) = \prod_{i=1}^{n} p(x_i|\theta),$$

with $p(x_i|\theta) = \theta^{x_i}(1 - \theta)^{1-x_i}$.

Remarks $\{X_i\}_{i=1}^{\infty}$ is an IES iff a generative model

$$\Theta \sim F, \quad X_i \sim p(\cdot|\theta), \text{ iid for } i = 1, 2, 3, ... \text{ when } \Theta = \theta.$$

exists for some F and $p(\cdot|\theta)$. See Bernoulli example in (3).

If $F(\theta)$ is a CDF for a density $\pi(\theta)$ so $dF(\theta) = \pi(\theta)d\theta$ then

$$p(x_1, ..., x_n) = \int_0^1 \prod_{i=1}^{n} p(x_i|\theta)\pi(\theta)d\theta.$$

In general F is just some unknown distribution which puts probability mass $F(A)$ in some sigma-field $\mathcal{B}_\Omega$ of subsets of Ω.

The theorem extends to infinite exchangeable sequences of random vectors ($X_i \in \mathcal{X}$ continuous multivariate random variables).

$$\Pr(X_1 \in A_1, \ldots, X_n \in A_n) = \int_{\mathcal{P}(\mathcal{X})} \prod_{i=1}^{n} Q(A_i)\, dF(Q).$$

with $A_i \subseteq \mathcal{X}$ and $\mathcal{P}(\mathcal{X})$ the distributions on $\mathcal{X}$.

Outline proof (of deF): $IES \Leftarrow (*)$ from the permutation symmetry in $(*)$. For $IES \Rightarrow (*)$, take $1 \leq n \leq N < \infty$, let

$$S_n = X_1 + X_2 ... + X_n, \quad n = 1, 2, ...$$

and let r, s be two integers $0 \leq r \leq s \leq N$. By exchangeability,

$$\Pr(S_n = r) = \binom{n}{r} p(x_1, ..., x_n) \qquad (A)$$

for any $x_1, ..., x_n$ summing to r and the distribution of S_n given $S_N = s$ is hypergeometric,

$$\Pr(S_n = r | S_N = s) = \binom{s}{r} \binom{N-s}{n-r} \Big/ \binom{N}{n} \qquad (B)$$

since this is the probability to draw r 1's in n draws without replacement from an urn containing s 1's and $N - s$ 0's.

When $S_n = r$ the urn contains at least $n - r$ 0's so

$$\begin{aligned}\Pr(S_n = r) &= \sum_{s=r}^{N-(n-r)} \Pr(S_n = r | S_N = s) \Pr(S_N = s) \\ &= \sum_{s=r}^{N-(n-r)} \Pr(S_n = r | S_N/N = \theta(s)) \Pr(S_N/N = \theta(s))\end{aligned}$$

where $\theta(s) \equiv s/N$.

The random variable

$$\Theta_N = S_N/N$$

takes values $\Theta_N \in \{0, 1/N, 2/N, ..., 1\}$. It has CDF F_N with

$$\begin{aligned}F_N(\theta) &= \Pr(\Theta_N \leq \theta), \\ &= \sum_{s=0}^{N} \Pr(S_N = s) \mathbb{I}_{s/N \leq \theta}\end{aligned}$$

Now "differentiate the step function"

$$dF_N(\theta) = \sum_{s=0}^{N} \Pr(S_N = N\theta)\delta_{s/N}(d\theta)$$

and write $\Pr(S_n = r)$ as an integral

$$\begin{aligned}\Pr(S_n = r) &= \sum_{s=r}^{N-(n-r)} \Pr(S_n = r | S_N/N = \theta(s)) \Pr(S_N/N = \theta(s)) \\ &= \int_0^1 \Pr(S_n = r | S_N = N\theta) dF_N(\theta)\end{aligned}$$

where the hypergeometric function $\Pr(S_n = r | S_N = N\theta)$ in (B) is equal zero when $N\theta \notin \{r, r+1, \ldots, N-(n-r)\}$.

Exercise: check by substituting dF_N and doing the integral (to integrate a δ-function just substitute $\theta(s)$ for θ in the integrand).

We now assume two limiting results. The hypergeometric $\Pr(S_n = r|S_N = N\theta)$ converges uniformly to the binomial,

$$\sup_{0\leq s\leq N}\left|\binom{s}{r}\binom{N-s}{n-r}\Big/\binom{N}{n} - \binom{n}{r}\left(\frac{s}{N}\right)^r\left(1-\frac{s}{N}\right)^{n-r}\right| \to 0$$

as $N \to \infty$ with n, θ fixed. Taking a small sample from a very large population is the same with or without replacement.

Helly's Theorem (Feller (1966) *Probability Theory and Applications* II) "an infinite sequence of probability distributions F_N on a finite interval contains a convergent subsequence".

Here, a limit $F_N(\theta) \to F(\theta)$ on a subsequence of N-values,

$$\lim_{N\to\infty} \Pr(\Theta_N \leq \theta) = \Pr(\Theta \leq \theta) \qquad \text{with } \Theta \sim F,$$

exists at all continuity-points $\theta \in [0, 1]$ of F.

Assuming these two convergence results,

$$\begin{aligned}\Pr(S_n = r) &= \lim_{N\to\infty} \int_0^1 \Pr(S_n = r | S_N = N\theta) dF_N(\theta) \\ &= \binom{n}{r} \int_0^1 \theta^r (1-\theta)^{n-r} dF(\theta). \qquad (C)\end{aligned}$$

Equating (A) and (C),

$$p(x_1, ..., x_n) = \int_0^1 \theta^r (1-\theta)^{n-r} dF(\theta) \qquad (D)$$

which is the result of de Finetti.

The representation in terms of F is unique. A distribution on a bounded interval is uniquely determined by its moments, and (D) fixes all the moments $E(\theta^n), n = 1, 2, ...$ of F (take $r = n$ and $n = 1, 2, ...$) so F exists and is unique. [EOP]

Bayesian inference (existence of prior)

If the data $X_1 = x_1, ..., X_n = x_n$ come from an IES then

$$p(x_1, ..., x_n|\theta) = \prod_i p(x_i|\theta)$$

is the likelihood. The generative model is

$$\Theta \sim F \quad \text{and} \quad X_i \sim p(\cdot|\theta),\ i = 1, ..., n$$

so a "true" prior F exists.

We can't often elicit the true generative model. We do our best to write down π to match F. At least the random variable Θ we are trying to model exists.

A similar issue arises for the likelihood - we know a conditional distribution for $X|\theta$ exists and do our best to approximate it.

Bayesian inference: (posterior)

Suppose we have seen $x_1, ..., x_m$ and we wish to predict $x_{m+1}, ..., x_n$. The predictive distribution is

$$\begin{aligned} p(x_{m+1:n}|x_{1:m}) &= p(x_{1:n})/p(x_{1:m}) \\ &= \int p(x_{m+1:n}|\theta)\frac{p(x_{1:m}|\theta)dF(\theta)}{p(x_{1:m})}. \end{aligned}$$

We see that

$$dF(\theta|x_1, ..., x_m) \propto p(x_1, ..., x_m|\theta)dF(\theta)$$

is our updated prior for θ given $x_{1:m}$, ie, the posterior.

de Finetti tells us an "unknown true" posterior distribution exists.